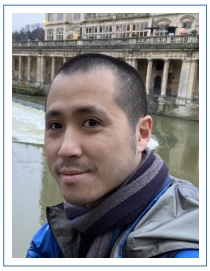
Toshiaki Asakura

Curriculum Vitae

Joint PhD candidate
Nagasaki University and
London School of Hygiene & Tropical Medicine
(+44) 7554 156529
toshiaki.asakura1@lshtm.ac.uk
My Webpage
Github *Linkedin* *ORCID*



Summary

I am a joint PhD candidate at Nagasaki University and the London School of Hygiene & Tropical Medicine, specialising in infectious disease epidemiology and mathematical modelling. My research revolves around Bayesian inference and transmision dynamic models to support public health decision-making, with particular interests in heterogeneity in social and sexual contact patterns, real-time outbreak response, and elimination/eradication of infectious diseases. Much of my recent work has focused on mpox (clades I and IIb) and contact patterns.

Looking ahead, I am motivated to generate scientific evidence hat directly informs policy - particularly for under-represented populations - and translate these insights into actionable outbreak responses. I am committed to producing open research and open source software development as means of fostering equitable access and collaborative science across diverse settings.

Education

2023–present **PhD, Infectious Disease Epidemiology and Modelling**, *Nagasaki University, London School of Hygiene & Tropical Medicine Joint PhD Degree Programme for Global Health*, Japan, UK, Supervisor: Dr Akira Endo and Dr Sebastian Funk .

2022–2023 : **MSc Control of Infectious Diseases**, *London School of Hygiene & Tropical Medicine*, UK, Dissertation: "A simulation-based assessment of the early detection ability of environmental surveillance for poliovirus" — Supervisor: Dr Kathleen M. O'Reilly.

2016–2022 : **Bachelor of Medicine**, *Hokkaido University*, Japan.

Publications

Published Journal Articles (* Corresponding author; † Equal contribution)

J13 Role of community and sexual contacts as drivers of MPXV clade I.
Hiroaki Murayama†, **Asakura, Toshiaki R.†**, Borame L. Dickens, Dorothy Boyle, Jen Han Foo, Shihui Jin, Patrick K. Mukadi, Keisuke Ejima, Sung-mok Jung, Akihiro Nishi, Kiesha Prem, Audry M. Wakamba, Diafuka Saila-Ngita, David Niyukuri, and Akira Endo.
Nature Health, pages 1–12, April 2026, doi:10.1038/s44360-026-00080-7.

J12 Disentangling temporal trends of Clade Ib monkeypox virus transmission in Burundi.
Shihui Jin, **Asakura, Toshiaki R.**, Hiroaki Murayama, Sung-mok Jung, David Niyukuri, Joseph Nyandwi, Liliane Nkengurutse, Olivier Kamatari, Jue Tao Lim, Akira Endo, and Borame L Dickens.
The Journal of Infectious Diseases, 233(1):e140–e144, January 2026, doi:10.1093/infdis/jiaf475.

J11 Vaccination strategies to achieve outbreak control for MPXV Clade I with a one-time mass campaign in sub-Saharan Africa: A scenario-based modelling study.
Shihui Jin, **Asakura, Toshiaki R.**, Hiroaki Murayama, David Niyukuri, Diafuka Saila-Ngita, Jue Tao Lim, Akira Endo, and Borame L. Dickens.
PLOS Medicine, 22(9):e1004726, September 2025, doi:10.1371/journal.pmed.1004726.

J10 Modelling international spread of clade IIb mpox on the Asian continent.
Asakura, Toshiaki R., Sung-Mok Jung, Hiroaki Murayama, Cyrus Ghaznavi, Haruka Sakamoto, Ayaka Teshima, Fuminari Miura, and Akira Endo.
Bulletin of the World Health Organization, 103(7):429–436, July 2025, doi:10.2471/BLT.24.291815.

J9 Assessing early detection ability through spatial arrangements in environmental surveillance for poliovirus: A simulation-based study.
Asakura, Toshiaki R.* and Kathleen M. O'Reilly.
PLOS ONE, 20(7):e0325789, July 2025, doi:10.1371/journal.pone.0325789.

J8 Characterising global risk profiles of Mpox clade Ib importation.
Asakura, Toshiaki R., Sung-mok Jung, Shihui Jin, Gang Hu, Akira Endo, and Borame Lee Dickens.
Journal of Travel Medicine, 31(8):taae136, December 2024, doi:10.1093/jtm/taae136.

J7 Traditional and social media usage associated with COVID-19 vaccine uptake in Sapporo, Japan.
Satoshi Sunohara, **Asakura, Toshiaki R.**, Takashi Kimura, Masayuki Saijo, and Akiko Tamakoshi.
Asia Pacific Journal of Public Health, 36(4):358–365, May 2024, doi:10.1177/10105395241240952.

J6 Patient characteristics and public health office factors associated with long reporting delay of COVID-19 cases in Sapporo City, Japan.
Daichi Watanuki, Akiko Tamakoshi, Takashi Kimura, **Asakura, Toshiaki**, and Masayuki Saijo.
Journal of Epidemiology, 34(3):129–136, 2024, doi:10.2188/jea.JE20220359.

J5 The association between COVID-19-related discrimination and probable post-traumatic stress disorder among patients with COVID-19 in Sapporo, Japan.
Isaku Kurotori, **Asakura, Toshiaki R.**, Takashi Kimura, Miyuki Hori, Mariko Hosozawa, Masayuki Saijo, Hiroyasu Iso, and Akiko Tamakoshi.
Journal of Epidemiology, 34(12):570–576, 2024, doi:10.2188/jea.JE20230360.

J4 Case–control study of Long COVID, Sapporo, Japan.
Asakura, Toshiaki†, Takashi† Kimura, Isaku Kurotori, Katabami Kenichi, Miyuki Hori, Mariko Hosogawa, Masayuki Saijo, Kaori Nakanishi, Hiroyasu Iso, and Akiko Tamakoshi.
Emerging Infectious Diseases, 29(5), May 2023, doi:10.3201/eid2905.221349.

J3 Relationship between out-of-facility deliveries and distance and travel time to delivery facilities in Hokkaido, Japan: An ecological study.
Yoshihiro Saito, **Asakura, Toshiaki**, Kimura Takashi, Takeshi Umazume, Hidemichi Watari, and Akiko Tamakoshi.
Journal of Obstetrics and Gynaecology Research, 49(3):930–937, 2023, doi:10.1111/jog.15543.

J2 Effective vaccine allocation strategies, balancing economy with infection control against COVID-19 in Japan.
Satoshi† Sunohara, **Asakura, Toshiaki†**, Takashi Kimura, Shun Ozawa, Satoshi Oshima, Daigo Yamauchi, and Akiko Tamakoshi.
PLOS ONE, 16(9):e0257107, September 2021, doi:10.1371/journal.pone.0257107.

J1 Behavioral characteristics of dopamine D5 receptor knockout mice.
Hitomi† Sasamori, **Asakura, Toshiaki†**, Chiaki Sugiura, Youcef Bouchekioua, Naoya Nishitani, Masaaki Sato, Takayuki Yoshida, Miwako Yamasaki, Akira Terao, Masahiko Watanabe, Yu Ohmura, and Mitsuhiro Yoshioka.
Scientific Reports, 12(1):6014, April 2022, doi:10.1038/s41598-022-10013-5.

Conference presentations and invited talks

C6 Poster presentation: Survey design matters for capturingheterogeneity in social contacts:from exponential to scale-free.
Asakura, Toshiaki R., Sebastian Funk, and Akira Endo.
In *10th Epidemics International Conference on Infectious Disease Dynamics*. San Diego, US., December 2025.

C5 Invited talk: Characterising global risk profiles of mpox clade ib importation.
Asakura, Toshiaki R., Sung-mok Jung, Shihui Jin, Gang Hu, Akira Endo, and Borame Lee Dickens.
In *Mpox Community Call at WHO Collaboratry*, October 2024.

C4 Poster presentation: Projecting international mpox spread in asia: ongoing global health risk.
Asakura, Toshiaki R., Sung-mok Jung, Hiroaki Murayama, Miura Fuminari, and Endo Akira.
In *The 34th Annual Scientific Meeting of the Japan Epidemiological Association*. Shiga, Japan, January 2024.

C3 Oral presentation: A quantitative simulation-based evaluation of the early detection of poliovirus using environmental surveillance.
Asakura, Toshiaki R. and Kathleen M. O'Reilly.
In *9th Epidemics International Conference on Infectious Disease Dynamics*. Bologna, Italy, December 2023.

C2 Oral presentation: The student's enlightenment activity against covid-19: No more corona project.
Asakura, Toshiaki R.
In *The 79th annual meeting of the Japanese Society of Public Health*. Kyoto, Japan, October 2020.

C1 Poster presentation: Monitoring the transmissibility of highly pathogenic avian influenza (h5n1) virus in poultry: A birth-death skyline model.
Asakura, Toshiaki R. and Hiroshi Nishiura.
In *World-leading next generation MD researchers and training project 2019 National Retreat*. Nagoya, Japan, April 2019.

Grants, Awards & Scholarships

2024 Japan Society for the Promotion of Science (JSPS) DC1 Research Fellowships and JSPS KAKENHI (Grants-in-Aid for Research Fellow of the JSPS), no. 24KJ1827. Amount: 1,400,000 (JPY)

2023 British Council Japan Association Scholar Award

2023 MSc Projects Funding, Trust Funds, London School of Hygiene & Tropical Medicine

2022 The Rotary Foundation. Global grant scholarship

2022 Hokkaido University, School of Medicine, Alumni Association President's Award

Academic service

Peer-review experiences: PLoS Computational Biology (1), VeriXiv (1), Nature Communication (1), Scientific Reports (2), BMC Public Health (1), BMC Infectious Diseases (2), Critical Public Health (1).

Teaching experience

Cosupervision: 1 MSc student (2025)

Teaching assistant: Introduction to Infectious Disease Modelling and its Applications, London School of Hygiene & Tropical Medicine (2025)
Introduction to infectious disease epidemiology and modelling, School of Tropical Medicine and Global Health, Nagasaki University (2023, 2024, 2025)

Media & Public engagement

March - May 2020 — Scientific communication about basic countermeasures against COVID-19 for young populations

Research experience

April - August 2022 — Academic Fellow, Faculty of Medicine, Hokkaido University - Supervisor: Akiko Tamakoshi

Skills

Academic skills	Mathematical modelling (compartment model, meta-population model, network-based model) Theoretical epidemiology (infectious disease epidemiology, cohort study) Computing statistics (Markov-Chain Monte Carlo method, Bayesian statistics, Gaussian process) Forecasting (scoring rules, forecasting evaluation)
Programming Languages	Julia, R, Python, SAS, C++
IT skills	Claude code, Docker, LaTex, Git/GitHub, Typescript, Django, React, Nginx
Language	Japanese (native), English (business), Chinese (daily)
Interests	Running, Wine, Travel, Novels